EX NO: 7 a

IMPLEMENTING REAL-TIME/TECHNICAL APPLICATIONS USING FILE HANDLING - COPY FROM ONE FILE TO ANOTHER

```
# Creating an output file in writing mode
output_file = open("OUTPUT.txt",'w')
# Opening input file and scanning each line
# from input file and writing in output file
with open("INPUT.txt",'r') as scan:
    output_file.write(scan.read())
    # Closing the output file
    output_file.close()
```

OUTPUT:





EX NO: 7 b

IMPLEMENTING REAL-TIME/TECHNICAL APPLICATIONS USING FILE HANDLING WORD COUNT

```
number_of_words = 0
with open(r"INPUT.txt",'r') as file:
# Reading the content of the file
# using the read() function and storing
# them in a new variable
data = file.read()
# Splitting the data into separate lines
# using the split() function
lines = data.split()
# Adding the length of the
# lines in our number_of_words
# variable
        number_of_words += len(lines)
# Printing total number of words
print(number_of_words)
```

OUTPUT:

```
5
```

EX NO: 8 a

IMPLEMENTING REAL-TIME/TECHNICAL APPLICATIONS USING EXCEPTIONHANDLING - DIVIDE BY ZERO ERROR

```
a=int(input("Entre a="))
b=int(input("Entre b="))
try:
   c = ((a+b) / (a-b))
   if a==b:
      raise ZeroDivisionError #Handling of error
   except ZeroDivisionError:
      print ("a/b result in 0")
   else:
      print (c)
```

OUTPUT:

```
Enter a=6
Enter b=4
5.0
```

EX NO: 8 b

IMPLEMENTING REAL-TIME/TECHNICAL APPLICATIONS USING EXCEPTIONHANDLING - CHECK VOTERS ELIGIBILITY

```
age=int(input("Enter your age "))
try:
    if age>18:
        print("Eligible to vote")
    else:
        print("Not eligible to vote")
except:
        print("age must be a valid number")
main()
```

OUTPUT:

```
Enter your
age 20
Eligible to
vote
```

EX NO: 9 a
IMPLEMENTING PROGRAMS USING WRITTEN MODULES AND PYTHON STANDARD LIBRARIES – PANDAS

```
import pandas as pd
df = pd.DataFrame({"Name": [ "Mr. Owen Harris", "Mr. William Henry","Miss. Elizabeth",],
"Age": [22, 35, 58], "Sex": ["male", "male", "female"],})
print(df)
print(df["Age"])
ages = pd.Series([22, 35, 58],name="Age")
print(ages)
df["Age"].max()
print(ages.max())
print(df.describe())
```

OUTPUT:

```
Name Age      Sex
0       Braund, Mr. Owen Harris 22  male
1       Allen, Mr. William Henry 35 male
2       Bonnell, Miss. Elizabeth 58 female
0       22
1       35
2       58
Name: Age, dtype: int64
0       22
1       35
2       58

Name: Age, dtype: int64 58
Age
count 3.000000
mean 38.333333
std     18.230012
min     22.000000
25%     28.500000
50%     35.000000
75%     46.500000
max     58.000000
```

EX NO: 9 b

IMPLEMENTING PROGRAMS USING WRITTEN MODULES AND PYTHON STANDARD LIBRARIES – NUMPY

```
import numpy as np
a = np.arange(6)
a2 = a[np.newaxis, :]
a2.shape
#Array Creation and functions:
a = np.array([1, 2, 3, 4, 5, 6])
a = np.array([[1, 2, 3, 4], [5, 6, 7, 8], [9, 10, 11, 12]])
print(a[0])
print(a[1])
np.zeros(2)
np.ones(2)
np.arange(4)
np.arange(2, 9, 2)
np.linspace(0, 10, num=5)
x = np.ones(2, dtype=np.int64)
print(x)
arr = np.array([2, 1, 5, 3, 7, 4, 6, 8])
np.sort(arr)
a = np.array([1, 2, 3, 4])
b = np.array([5, 6, 7, 8])
np.concatenate((a, b)) #Array Dimensions:
array_example = np.array([[[0, 1, 2, 3], [4, 5, 6, 7]], [[0, 1, 2, 3], [4, 5, 6, 7]],[[0 ,1 ,2, 3], [4, 5, 6, 7]]])
array_example.ndim
array_example.size
array_example.shape
a = np.arange(6)
print(a)
b=a.reshape(3, 2)
print(b)
np.reshape(a, newshape=(1, 6), order='C')
```

OUTPUT:

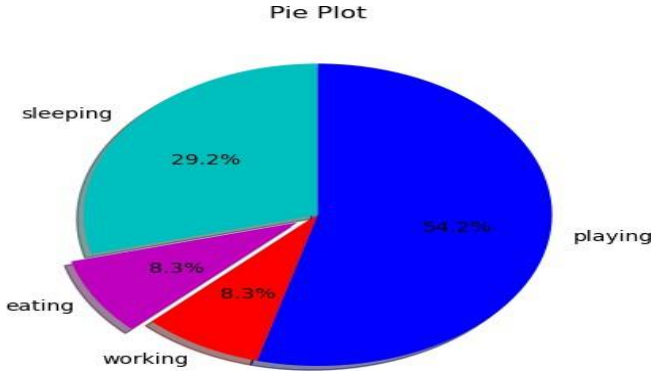
```
[1 2 3 4]
[5 6 7 8]
[1 1]
[0 1 2 3 4 5]
[[0 1]
[2 3]
[4 5]]
array([[0, 1, 2, 3, 4, 5]])
```

EX NO: 9 c

IMPLEMENTING PROGRAMS USING WRITTEN MODULES AND PYTHON STANDARD LIBRARIES– MATPLOTLIB

```
import matplotlib.pyplot as plt
days = [1,2,3,4,5]
sleeping =[7,8,6,11,7]
eating = [2,3,4,3,2]
working =[7,8,7,2,2]
playing = [8,5,7,8,13]
slices = [7,2,2,13]
activities = ['sleeping','eating','working','playing']
cols = ['c','m','r','b']
plt.pie(slices,labels=activities, colors=cols, startangle=90,shadow= True, explode=(0,0.1,0,0),
autopct='%1.1f%%')
plt.title('Pie Plot')
plt.show()
```

OUTPUT:

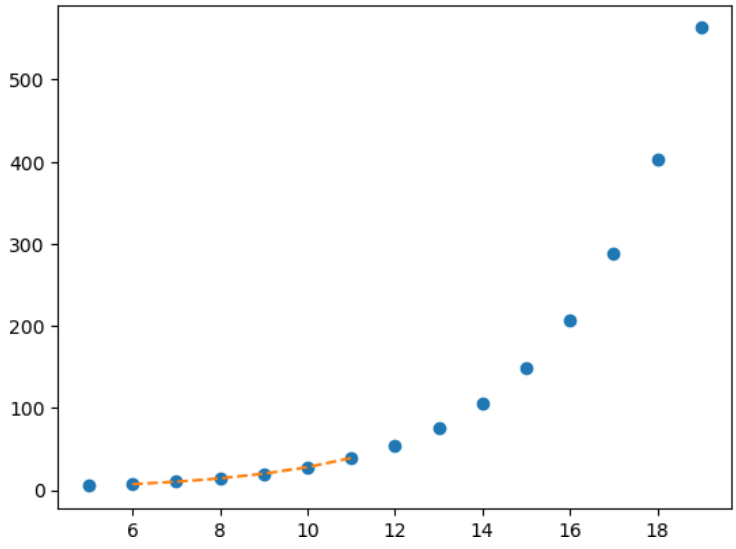


EX NO: 9 d

IMPLEMENTING PROGRAMS USING WRITTEN MODULES AND PYTHON STANDARD LIBRARIES-SCIPY

```
import matplotlib.pyplot as plt
from scipy import interpolate
import numpy as np
x = np.arange(5, 20)
y = np.exp(x/3.0)
f = interpolate.interp1d(x, y)
 x1 = np.arange(6, 12)
y1 = f(x1) # use interpolation function returned by `interp1d`
plt.plot(x, y, 'o', x1, y1, '--')
plt.show()
```

Output



EX NO: 10 a

IMPORT A DATASET AND EXPLOREIT USING PANDAS PACKAGE

```
import pandas as pd
data = {'animal': ['cat', 'cat', 'snake', 'dog', 'dog', 'cat', 'snake', 'cat', 'dog', 'dog'],
'age': [2.5, 3, 0.5, np.nan, 5, 2, 4.5, np.nan, 7, 3],
'visits': [1, 3, 2, 3, 2, 3, 1, 1, 2, 1],
'priority': ['yes', 'yes', 'no', 'yes', 'no', 'no', 'no', 'yes', 'no', 'no']}
labels = ['a', 'b', 'c', 'd', 'e', 'f', 'g', 'h', 'i', 'j']
df = pd.DataFrame(data, index=labels)
df
```

OUTPUT:

	animal	age	visits	priority
a	cat	2.5	1	yes
b	cat	3.0	3	yes
c	snake	0.5	2	no
d	dog	NaN	3	yes
e	dog	5.0	2	no
f	cat	2.0	3	no
g	snake	4.5	1	no
h	cat	NaN	1	yes
i	dog	7.0	2	no
j	dog	3.0	1	no

PROGRAM:

```
df.info()
```

OUTPUT:

```
<class 'pandas.core.frame.DataFrame'>
Index: 10 entries, a to j
Data columns (total 4 columns):
 #   Column    Non-Null Count  Dtype
---  ------    --------------  -----
 0   animal    10 non-null     object
 1   age       8 non-null      float64
 2   visits    10 non-null     int64
 3   priority  10 non-null     object
dtypes: float64(1), int64(1), object(2)
memory usage: 400.0+ bytes
```

PROGRAM:

```
df.head(3)
```

OUTPUT:

	animal	age	visits	priority
a	cat	2.5	1	yes
b	cat	3.0	3	yes
c	snake	0.5	2	no

PROGRAM:

df.loc[df.index[[3, 4, 8]], ['animal', 'age']]

OUTPUT:

	animal	age
d	dog	NaN
e	dog	5.0
i	dog	7.0

PROGRAM:

df['visits']>2

OUTPUT:

a False
b True
c False
d True
e False
f True
g False
h False

PROGRAM:

df[df['age'].isnull()]

OUTPUT:

	animal	age	visits	priority
d	dog	NaN	3	yes
h	cat	NaN	1	yes

PROGRAM:

df[(df['animal']=='cat')&(df['age']<3)]

OUTPUT:

	animal	age	visits	priority
a	cat	2.5	1	yes
f	cat	2.0	3	no

PROGRAM:

df[df['age'].between(2,4)]

OUTPUT:

	animal	age	visits	priority
a	cat	2.5	1	yes
b	cat	3.0	3	yes
f	cat	2.0	3	no
j	dog	3.0	1	no

PROGRAM:

```
df.groupby('animal')['age'].mean()
```

OUTPUT:

```
animal
cat      2.5
dog      5.0
snake    2.5
Name: age, dtype: float64
```

PROGRAM:

```
df['animal'].value_counts()
```

OUTPUT:

```
animal
```

PROGRAM:

```
df.sort_values(by=['age', 'visits'], ascending=[False,True])
```

OUTPUT:

	animal	age	visits	priority
i	dog	7.0	2	no
e	dog	5.0	2	no
g	snake	4.5	1	no
j	dog	3.0	1	no
b	cat	3.0	3	yes
a	cat	2.5	1	yes
f	cat	2.0	3	no
c	snake	0.5	2	no
h	cat	NaN	1	yes
d	dog	NaN	3	yes

EX NO: 10 b

IMPORT A DATASET AND EXPLOREIT USING NUMPY PACKAGE

```
import numpy as np
L = list(range(10))
[str(c) for c in L]
```

OUTPUT

```
['0', '1', '2', '3', '4', '5', '6', '7', '8', '9']
```

PROGRAM

```
[type(item) for item in L]
```

OUTPUT

```
[int, int, int, int, int, int, int, int, int, int]
```

PROGRAM

```
np.zeros(10, dtype='int')
```

OUTPUT

```
array([0, 0, 0, 0, 0, 0, 0, 0, 0, 0])
```

PROGRAM

```
np.ones((3,5), dtype=float)
```

OUTPUT:

```
array([[ 1.,  1.,  1.,  1.,  1.],
       [ 1.,  1.,  1.,  1.,  1.],
       [ 1.,  1.,  1.,  1.,  1.]])
```

PROGRAM

```
np.full((3,5),1.23)
```

Output

```
array([[ 1.23,  1.23,  1.23,  1.23,  1.23],
       [ 1.23,  1.23,  1.23,  1.23,  1.23],
       [ 1.23,  1.23,  1.23,  1.23,  1.23]])
```

PROGRAM

```
np.arange(0, 20, 2)
```

OUTPUT

```
array([0, 2, 4, 6, 8,10,12,14,16,18])
```

PROGRAM

```
np.linspace(0, 1, 5)
```

OUTPUT

```
array([ 0., 0.25, 0.5 , 0.75, 1.])
```

PROGRAM

```
np.random.normal(0, 1, (3,3))
```

OUTPUT

```
array([[ 0.72432142, -0.90024075,  0.27363808],
       [ 0.88426129,  1.45096856, -1.03547109],
       [-0.42930994, -1.02284441, -1.59753603]])
```

PROGRAM

```
np.eye(3)
```

OUTPUT

```
array([[ 1.,  0.,  0.],
       [ 0.,  1.,  0.],
```

[0., 0., 1.]])

PROGRAM

```
np.random.seed(0)
x1 = np.random.randint(10, size=6)
#one dimension
x2 = np.random.randint(10, size=(3,4))
#two dimension
x3 = np.random.randint(10, size=(3,4,5))
#three dimension
print("x3 ndim:", x3.ndim)
print("x3 shape:", x3.shape)
print("x3 size: ", x3.size)
```

OUTPUT

```
('x3 ndim:', 3)
('x3 shape:', (3, 4, 5))
('x3 size: ', 60)
```